Sean Luka Girgis

Senior Data Engineer | AWS, ETL Validation, Python, PySpark & Data Quality

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Professional Summary

Senior Data Engineer with 20+ years of enterprise technology experience across AWS data platforms, Python, SQL, PySpark/Spark, Redshift, Glue, S3, ETL/ELT pipelines, data validation, reconciliation, automated quality checks, production troubleshooting, and data warehouse reliability. Strong background building and validating high-volume data workflows, optimizing SQL, documenting transformation logic, investigating pipeline failures, and supporting governed reporting datasets for regulated enterprise environments. Best aligned to data engineering QA, ETL validation, data quality, and pipeline reliability roles requiring AWS Redshift/Glue, Python automation, PySpark, reconciliation, test coverage, and cross-functional delivery; dbt, CDK, DMS, Deequ, Step Functions, SNS/SQS, EventBridge, Node.js, and GenAI are positioned carefully as adjacent or active-learning areas where appropriate.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark ETL pipelines ingesting telemetry from 6,000+ infrastructure endpoints and transforming raw operational data into curated datasets for reporting, analytics, forecasting, and validation workflows.
- Designed AWS-based data platform components using S3 landing zones, AWS Glue, Athena, Redshift, Oracle, SQL, and PySpark to support scalable transformation, querying, and stakeholder-facing reporting.
- Built data validation, reconciliation, and quality review practices across BMC TrueSight, CA Wily, AppDynamics, Oracle, and AWS-backed reporting layers to improve accuracy, completeness, and trust in pipeline outputs.
- Developed SQL and Python validation logic to compare source feeds, transformation outputs, historical trends, and reporting-layer results for data quality and regression issues.
- Investigated production data failures by tracing source extraction, transformation logic, SQL behavior, failed outputs, schema drift indicators, and downstream reporting discrepancies.
- Optimized SQL queries, Oracle schemas, Redshift-backed workloads, indexes, views, partitioning patterns, and historical retention models for performance, maintainability, and cost-aware access.
- Documented transformation rules, test assumptions, validation checkpoints, metric definitions, operational runbooks, and support steps to improve repeatability and auditability.
- Partnered with infrastructure, analytics, BI-style reporting, application, and business teams to identify testing impacts, mitigate data risks, and deliver reliable data products.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Developed analytics and reporting outputs from Dynatrace AppMon and Gomez Synthetic Monitoring data for Brand.com and critical hospitality systems.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and communicate findings to engineering teams.

- Built dashboards and operational reporting views that converted raw monitoring data into actionable performance and reliability insights.
- Supported production issue analysis by correlating monitoring signals, transaction behavior, and system performance data.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Managed enterprise telemetry environments with 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting monitoring, operational analytics, dashboards, and production reliability.
- Built Perl and Korn shell automation for extracting, transforming, and distributing performance data, replacing manual exports with repeatable reporting workflows.
- Designed dashboards, alert logic, SLA thresholds, technical documentation, and reusable reporting standards adopted across client environments.
- Partnered with J2EE, WebLogic, infrastructure, database, and operations teams to analyze data, troubleshoot bottlenecks, and resolve production issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed enterprise telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, CPU, memory, thread, and garbage-collection constraints.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance and regression testing.
- Documented baseline metrics, test findings, regression thresholds, and release-readiness notes used by engineering teams for production decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster, preserving transaction performance and data integrity.
- Designed and optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Automated schema conversion, data validation, and phased cutover support for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation and integration patterns for structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces and reporting processes for telecom-scale transaction systems.

- Worked with XML, SQL, Oracle, PL/SQL, messaging, and integration interfaces supporting billing, customer-management, and operational reporting workflows.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Built automation scripts in Korn shell, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, and operational housekeeping.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, XML, JSON, Perl, Korn Shell, C++, Java

Flagship Projects

Serverless Lakehouse Platform (AWS)
Designed a scalable AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, query optimization, reporting access, and validation checkpoints.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, SQL

HorizonScale — AI Capacity Forecasting Engine
Built a forecasting-oriented analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure telemetry, prepare model-ready datasets, validate pipeline outputs, and deliver capacity insights.
Technologies: Python, PySpark, SQL, Prophet, scikit-learn, Streamlit, Data Validation

AI-Powered Job Search Pipeline
Built an AI-assisted Python automation pipeline with semantic duplicate detection, LLM-based scoring, JSON artifact generation, quality gates, document rendering, and application tracking using Claude Code, Grok/xAI APIs, FAISS, and sentence-transformers.
Technologies: Python, Claude Code, Grok / xAI API, FAISS, Sentence Transformers, JSON, Quality Gates